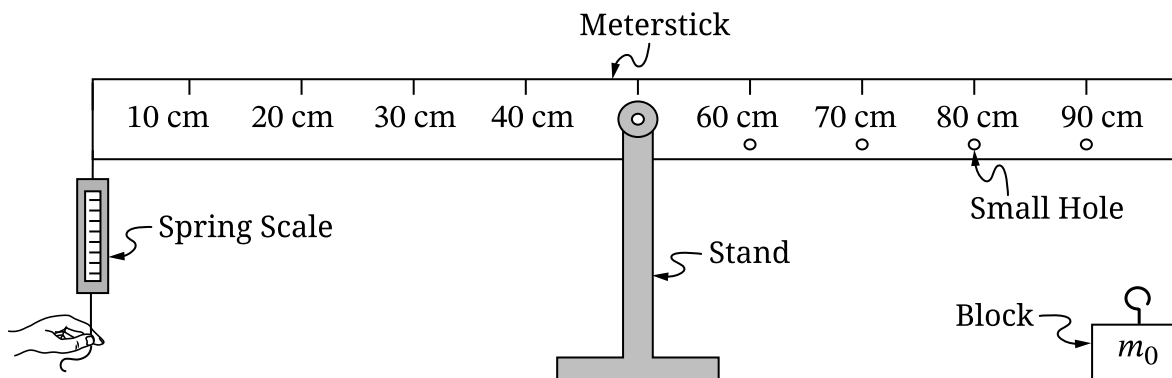


Question 3: Version J

3. Students are investigating balancing systems using the following setup. The students have a spring scale of negligible mass that is fixed to one end of a uniform meterstick. The center of the meterstick is attached to a stand on which the meterstick can pivot. There is a hook of negligible mass fixed to the top of a block of mass m_0 . The hook can be attached to the meterstick through one of the small holes in the meterstick, as shown in Figure 1. The students do not have a direct way to measure the mass of the block. The block cannot be attached to the spring scale.

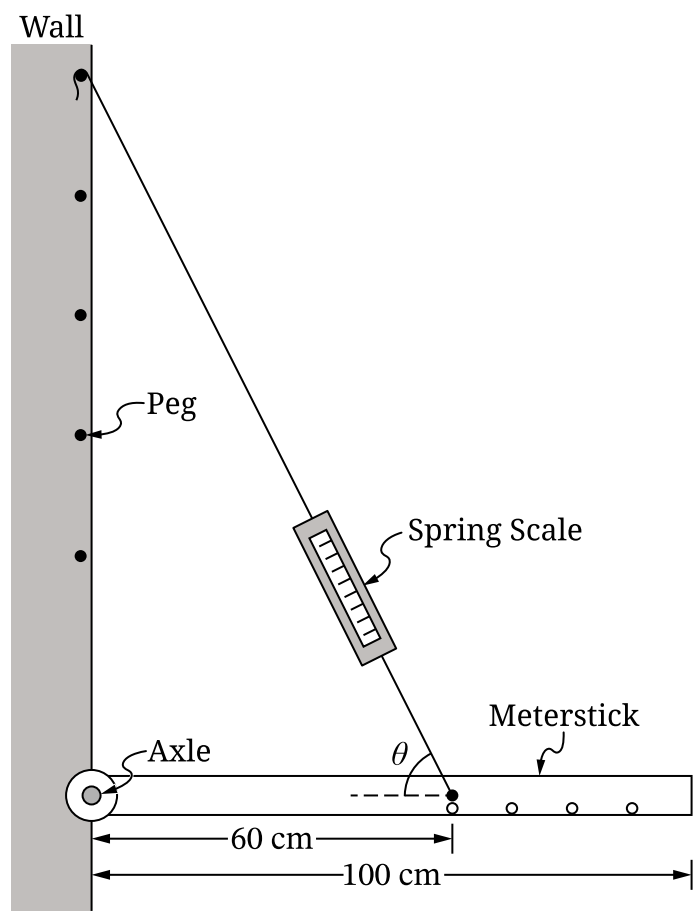
Figure 1

The students are asked to take measurements that will allow the students to create a linear graph whose slope could be used to determine the mass m_0 of the block.

- A. Describe** an experimental procedure to collect data that would allow the students to determine m_0 . Include any steps necessary to reduce experimental uncertainty.
- B. Describe** how the data collected in part A could be graphed and how that graph would be analyzed to determine m_0 .

The students have an identical meterstick of mass M that is now attached to an axle that is fixed to a wall. The meterstick is free to rotate with negligible friction about the axle. The meterstick is suspended horizontally by a string that is connected to a spring scale of negligible mass, as shown in Figure 2.

Figure 2



The angle θ that the string makes with the meterstick can be varied by attaching the string to one of the pegs located along the wall. The students use the spring scale to measure the tension F_T required to hold the meterstick horizontal. Table 1 shows the measured values of θ and F_T .

Table 1

θ (degrees)	F_T (N)
22	21
31	17
36	13
45	12
80	8

The students correctly determine that the relationship between F_T and θ is given by

$$F_T = \frac{5Mg}{6 \sin \theta}.$$

The students create a graph with $\frac{1}{\sin \theta}$ plotted on the horizontal axis.

C.

- i. **Indicate** what measured or calculated quantity could be plotted on the vertical axis to yield a linear graph whose slope can be used to calculate an experimental value for the mass M of the meterstick.

Vertical axis: _____ Horizontal axis: $\frac{1}{\sin \theta}$

- ii. On the blank grid provided, create a graph of the quantities indicated in part C (i) that can be used to determine M .

- Use Table 2 to record the data points or calculated quantities that you will plot.
- Clearly **label** the vertical axis, including units as appropriate.
- **Plot** the points you recorded in Table 2.

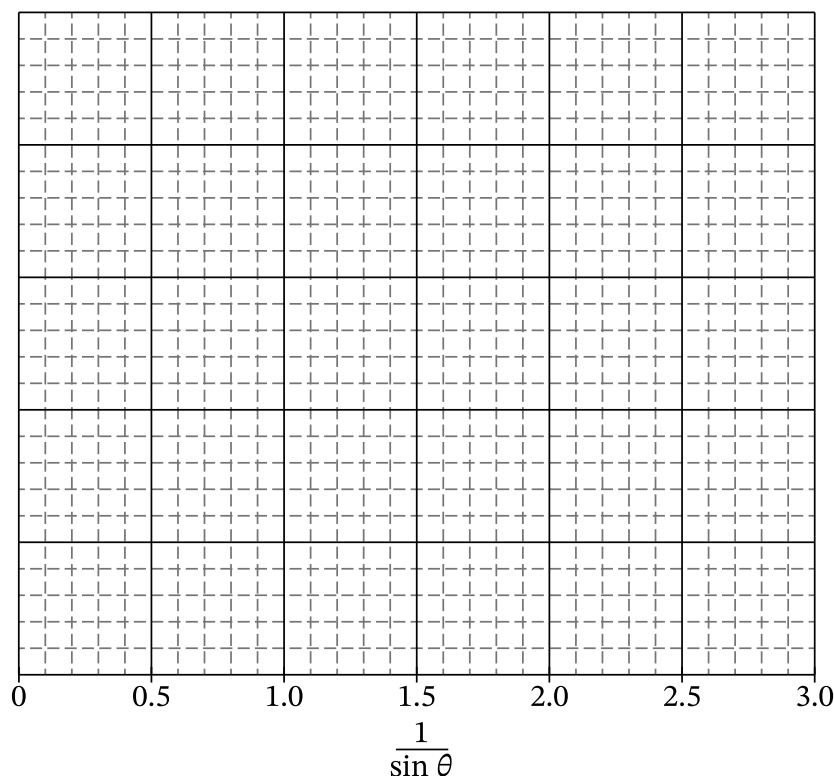


Figure 3

- iii. **Draw** a straight best-fit line for the data graphed in part C (ii).

- D. Using the best-fit line that you drew in part C (iii), **calculate** an experimental value for the mass M of the meterstick.